

“Many disciplines are now involved in getting a final electronic product out of the drawing board”



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This discussion takes a look at how electronics design tools have evolved and what new challenges these can now help solve. Elmar Dukek, EMEA VAR manager, and Sreekanth Tammana, country manager—India, Pakistan, Bangladesh and Sri Lanka, Altium Europe GmbH, speak with Dilin Anand from EFY

Q. Have you observed any interesting trends over the years on how the overall product design flow is evolving?

A. Many disciplines are now involved in getting a final electronic product out of the drawing board. You need electronics engineers for designing the modern printed circuit board, housing designers, and product developers for defining the form factor of the final product and other details that influence the end-user. You have hardware engineers who build the system. If it is an intelligent device or a smart device, like infotainment systems in cars, then you get embedded engineering and software engineers involved. Then, there is the Cloud system engineer who connects these devices to the Cloud and develops front-end interfaces to manage devices and grab data out of these.

Q. How important is 3D mechanical design capability for an electronics design engineer?

A. 3D integration is required because many design engineers previously faced problems with fitting the board

into the housing. They wanted to know how they could manage this in the design phase itself, as well as perform collision checking, apart from ensuring that all ports fit perfectly on the PCB, and so on. That is why it became important to interface properly with the mechanical world. If the tool has support for incremental design exchange format (IDX) files, it would help with the integration of design files from electronics design automation (EDA) tools to mechanical design tools, like SolidWorks. The benefit here is that IDX allows incremental changes to the design file, in addition to making the file sizes smaller, as compared to standards like IDF or STEP.

Q. How can an engineer help improve the design experience, making it relatively hassle-free?

A. Having easy access to data management capabilities of engineering design data and integrated libraries, a proper server for managing components and design for re-use stuff, and being able to manage the output of design data would be a great help, as these make the design process much more straightforward and robust.

Another feature is the partnership on the content side done by EDA tool vendors. Here, the designer gets access to important data relating to pricing information, stock levels, risk of obsolete components and more from within the design tool during the process.

Q. Please elaborate on the benefits of data management?

A. In the usual process of data management, designers have a common shared folder on the network drive and everybody puts their data on it. They modify, rename the data in the folder, delete or move it around as and when it suits them. Version control and revision management is often done on the basis of different folder names. This results in a situation where you cannot guarantee proper quality of data. By using a central repository, every single change made to a design can be tracked to the exact file.

Q. What are the latest features that design engineers are looking out for in EDA tools?

A. Design engineers, who are our customers, look for better routing solutions, high-speed and high-frequency support. They also look for help in handling complex processes, like tuning impedances and the length of different leads. Another new feature they find useful is pin-pair functionality, which means that a user can define a pin pair with different lengths and then tune it accordingly. ●